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Lesson 10: Convolutional Networks and Course Synthesis

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

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Agenda

- A convolution, and the two assumptions inside it
- Equivariance and pooling: how position stops mattering
- One defect, moved half an image
- Transfer learning, and its narrow window
- What ten lessons add up to

Exercise 9, before we start

- What counts is the **spread** reported beside each number, not the number
- The gap to watch for: a hand-written backward pass, never gradient-checked
- Conclusions drawn from a single seed, presented as differences
- Today the same obligation lands on an **architecture** claim

The sentence lesson 9 ended on

- “Permute the pixels consistently and nothing here notices”
- True of every model in that lesson, and an admission
- A 24×24 image is **not** 576 unrelated numbers
- Neighbouring pixels are the whole point of a photograph
- Today: what you get by refusing to throw the arrangement away

One assumption changes

- Same loss, same optimiser, same backpropagation as last week
- One new layer type, and inside it **two** assumptions
- Everything else today is consequence, or measurement
- No new mathematics after the first twenty-five minutes

Not “convolution wins on images”

- A random forest on raw pixels will come within **0.002** today
- Lesson 9’s dense network will lose to plain logistic regression
- Four hand-written numbers put every classical family at the ceiling
- What wins is the **representation**, not the model family

Three results that contradict a slogan

- Permuting the pixels costs the convolution **nothing** on detection
- Transfer learning helps in a **window**, and is negative outside it
- Freezing a pre-trained base hurt even the **good** source
- All three are measured this afternoon, in front of you

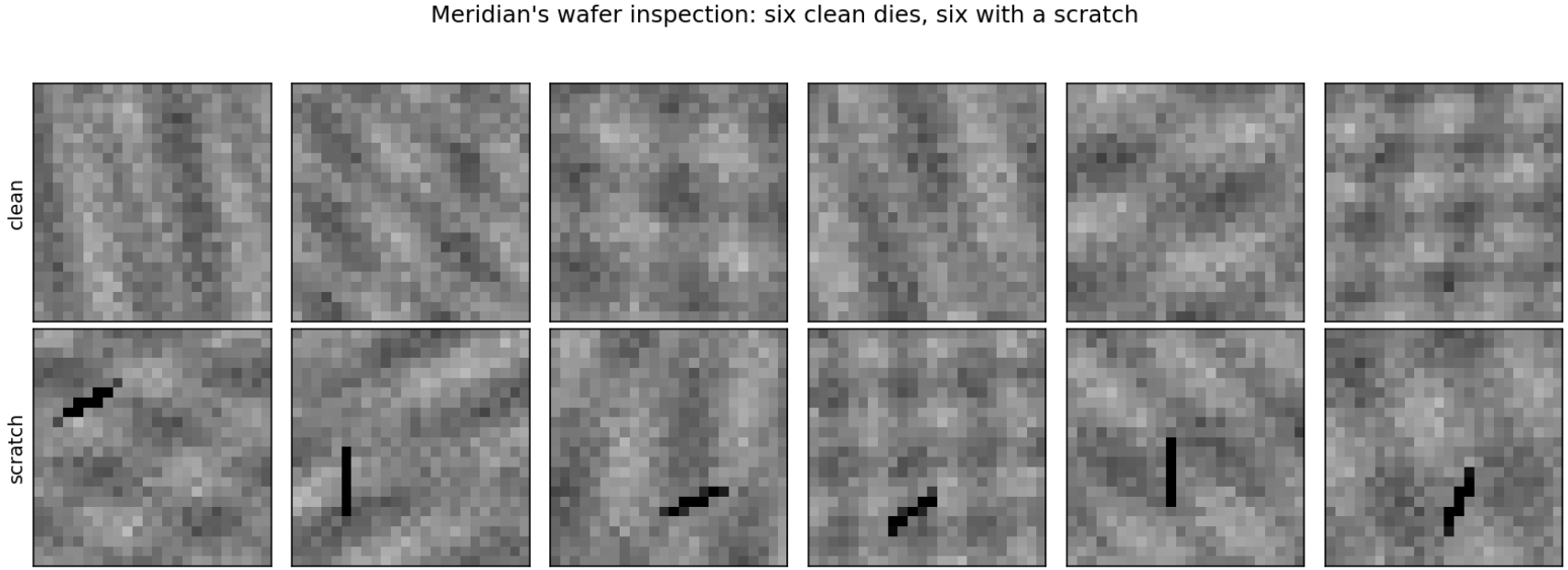
Five numbers so far; today the sixth

- **77%** accuracy on coin-flip labels, from leakage alone
- **94 of 128** imputed rows borrowed from the test set
- **365**: the largest coefficient a 0.01 penalty leaves
- **37 of 40** disguised accounts caught by reconstruction error
- A sigmoid layer divides the gradient by about **four**

Meridian’s wafers

- Lesson 9’s optical sensors are cut from silicon wafers
- Every die is photographed at **24×24 pixels** and graded
- Rejected if it carries a defect: a scratch of about seven pixels, or a bright particle
- Film thickness varies smoothly across each die

Six clean dies, six scratched



Two properties decide everything

- **Local**: a handful of the 576 pixels matter, the rest is irrelevant
- **Position-independent**: top left and bottom right are the same event
- Both were visible in the twelve photographs
- Every design choice this afternoon follows from these two lines

The grading station is wrong 2% of the time

- Published error rate **2%**; notebook 01’s batch drew **2.55%**
- A high draw at 2.5 standard deviations, not a bug
- Every score today is read against the ceiling of the set it was measured on
- For the 2,000-image test set that ceiling is **0.9800**

One neuron per pixel is the wrong shape

- A dense unit on this image has **576 weights**, one per pixel
- A unit that has learned to spot a scratch here…
- …knows nothing about the same scratch three pixels to the left
- Those are different weights, trained by different examples
- Nothing in the architecture connects them

80 weights against 147,712

layer	parameters
dense, 256 units	147,712
dense, 1024 units	590,848
convolutional, 8 kernels of 3×3	**80**
convolutional, 32 kernels of 5×5	832

Compression is the least interesting reading

- The convolutional layer produces **more** numbers, not fewer
- Eight feature maps of 24×24 is **4,608** outputs, against 256
- It is not a smaller layer: it is a differently **tied** one
- The 1,846-fold parameter ratio is a side effect

What weight sharing buys is evidence

- The same nine weights are applied at **every** position
- So one scratch, anywhere, teaches the detector **everywhere**
- The dense layer must be taught position by position
- Not a memory saving: a **data** saving
- We price it, in images, at 1:35

Before the mathematics: what would you compute?

- Look at the twelve dies again
- What number would you compute from an image to grade it?
- If your answer involves a small window and the word “anywhere”…
- …you have already derived the next forty minutes

The convolution, in words

- Take a small array of weights: the **kernel**
- Slide it over the image, one position at a time
- At each position, write down the weighted sum of the pixels it covers
- The result is a **feature map**: one number per position
- The kernel is small; the image is not

One weighted sum, at every position

$$(I * K)[r, c] = \sum_{i=0}^{f-1} \sum_{j=0}^{f-1} I[r+i,\ c+j]\ K[i, j]$$

Strictly, this is cross-correlation

- A true convolution flips the kernel before multiplying
- Every deep learning library computes the formula above and calls it convolution
- The kernel is **learned**, so the flip changes nothing that matters
- Notebook 01 agrees with scipy to 8.9×10^{-16}: rounding, not error

How big is the output?

$$\text{output size} = \left\lfloor \frac{n + 2p - f}{s} \right\rfloor + 1$$

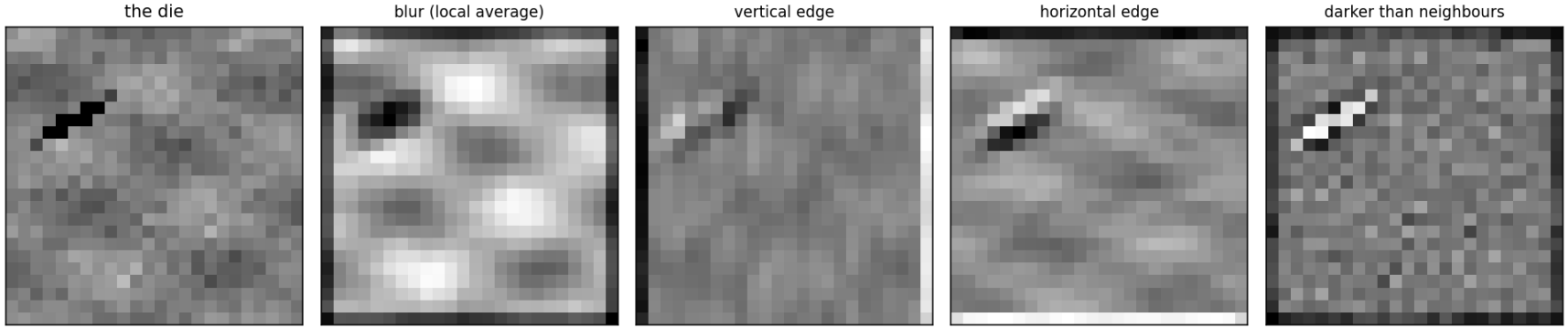
Worked on this lesson's 24×24 images

kernel	padding	stride	output
3	0	1	22
3	1	1	**24**
5	2	1	**24**
3	1	2	**12**

Two cases cover almost everything

- **3×3 with padding 1 leaves the size unchanged**: Keras calls this `padding="same"`
- In general, padding = (kernel − 1) / 2 for odd kernels
- **Stride 2 halves it**: the cheap alternative to pooling
- Almost every architecture you will read is built from these two

Four kernels written down, not learned



A constant patch gives exactly zero

$$\sum_{i,j} v \cdot K[i,j] = v \sum_{i,j} K[i,j] = v \cdot 0 = 0$$

Nine numbers, no training: 3.129 against 1.118

- Centre +8, each of the eight neighbours −1, sum exactly zero
- Blind to brightness, sensitive only to **local contrast**
- Strongest response on a defective die: **3.129**
- Strongest response on a clean die: **1.118**
- Nothing has been trained yet

Notebook 1, live

- The convolution in eleven lines, checked against scipy
- Output sizes measured, then compared with the formula
- Four kernels by hand, and the zero-sum one that works
- Equivariance checked at four offsets, then pooling

Break

- Twelve minutes

Shift the input, and the response shifts with it

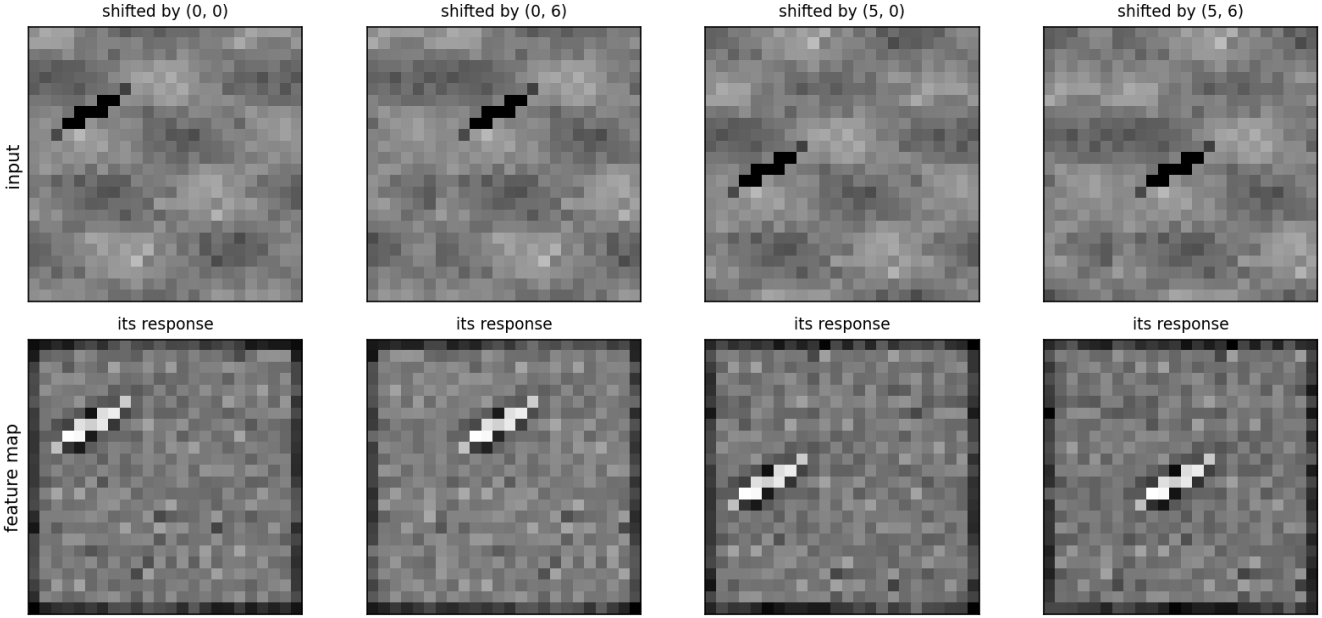
- **Equivariance**: the response does not change, it **moves**
- It follows in one line from the definition
- Notebook 01 checks it at four offsets: largest difference exactly **0**
- A dense layer has nothing of the kind
- Shift by one pixel and all 576 weights meet different pixels

Shift, then convolve, is convolve, then shift

$$(\text{shift}_d\ I) * K = \text{shift}_d(I * K)$$

The bright spot tracks the defect

move the defect and the bright spot moves with it, the detector does not have to be retrained

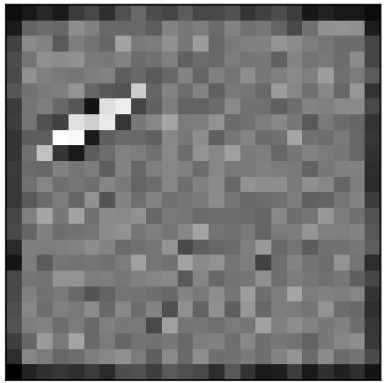


Pooling: from *where* to *whether*

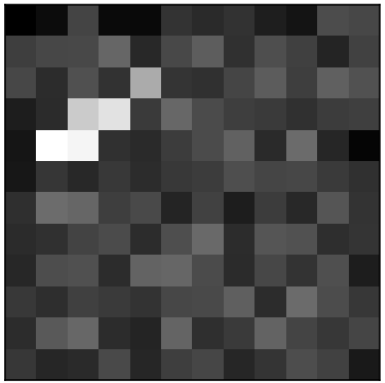
- Equivariance says the response **moves** with the defect
- To grade a die we do not want it to move
- We want one number: did a strong response occur **anywhere**?
- Max pooling takes the maximum over a window, discarding the position
- 24×24 down to 3×3, and the maximum is still 3.129

Pooled three times; the maximum never moves

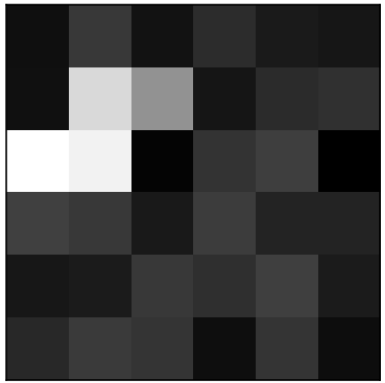
feature map
24x24



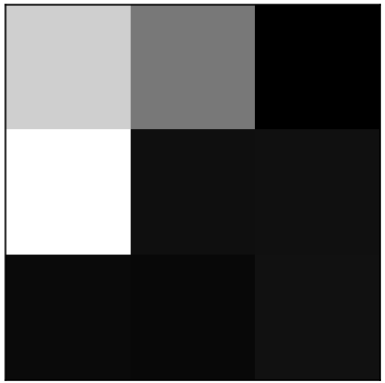
after 1 pooling step
12x12



after 2 pooling steps
6x6



after 3 pooling steps
3x3



Three things pooling does

- Turns equivariance into **invariance**: one pixel of shift often changes nothing
- Enlarges the receptive field: 3×3 after pooling covers 6×6 of the image
- Discards spatial precision, which is a **cost** when *where* is the answer
- **Global** max pooling: one number per kernel, over the whole map

1,537 parameters, in full

layer	output shape	parameters
conv, 8 kernels of 3×3, same	24 × 24 × 8	80
max pool 2×2	12 × 12 × 8	0
conv, 16 kernels of 3×3, same	12 × 12 × 16	1,168
max pool 2×2	6 × 6 × 16	0
global max pool	16	0
dense 16, then dense 1	1	289
total		**1,537**

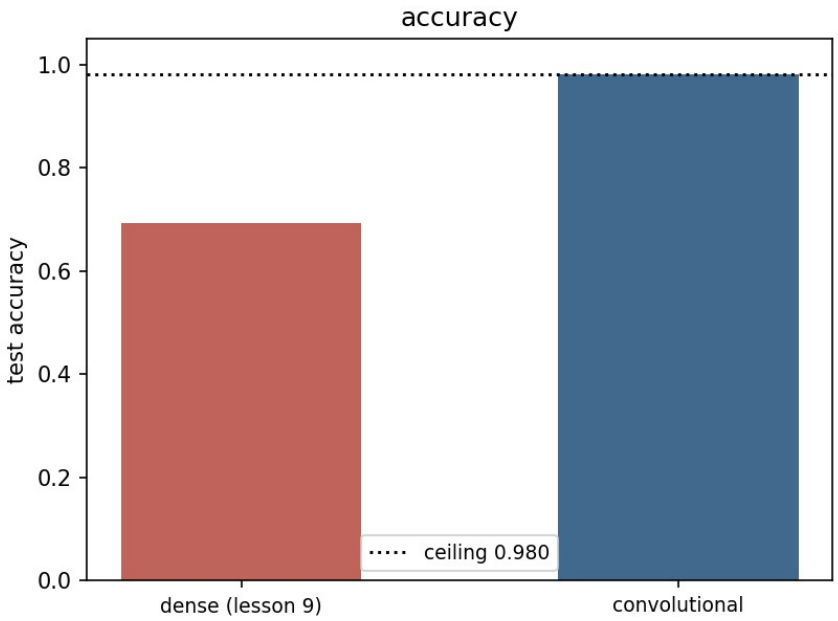
The count students get wrong: 1,168

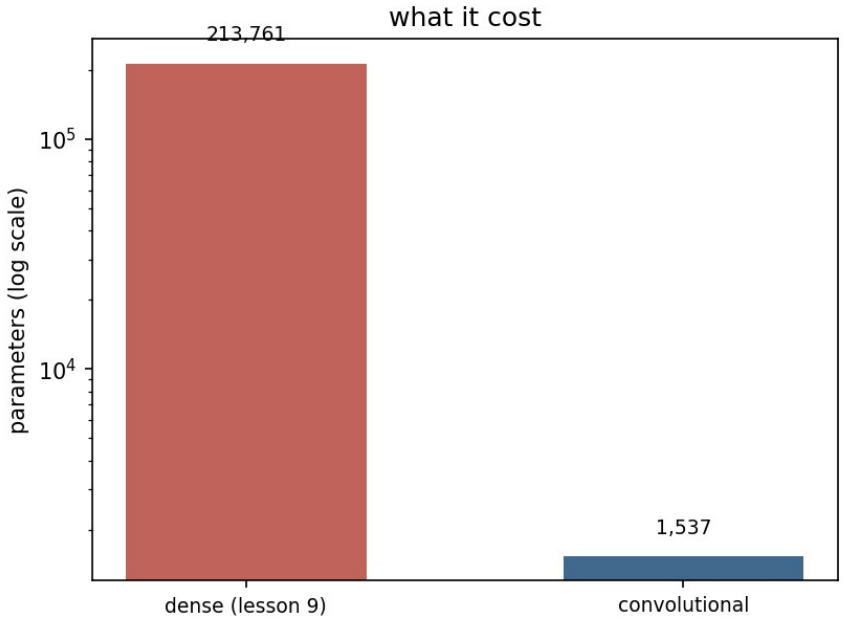
- Each of the 16 kernels is 3×3 **across all eight input channels**
- So each one holds 9 × 8 = 72 weights, not 9
- 16 × 72 + 16 = 1,168
- A kernel’s depth always matches the input’s channel count
- Only its two spatial dimensions are yours to choose

Head to head, and where the missing 2% lives

- Dense network, lesson 9: **0.6928**, standard deviation 0.0342
- Convolutional: **0.9800**, standard deviation 0.0000
- Against the **true** grade the convolutional network scores **1.0000**
- Its visible 0.98 is the grading station’s error and nothing else
- There is no gap left for a better architecture to close

1,537 parameters against 213,761





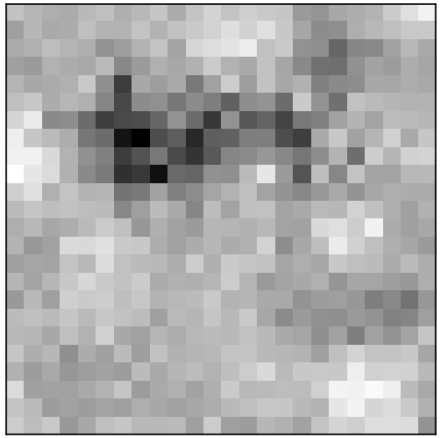
A defect where none has been seen

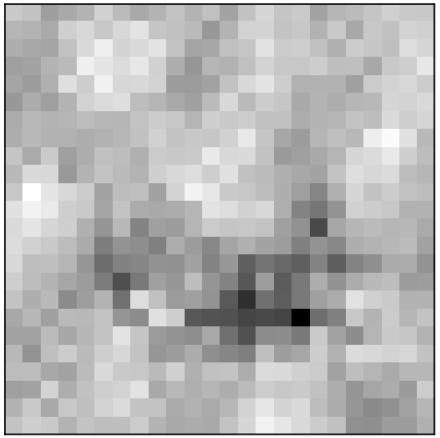
- Training defects live only in the **top** band of rows
- One test set keeps them there; the other moves them to the **bottom**
- Same generator, same defect, same contrast, same number of images
- Only the row changed
- Predict the two numbers before you see them

The number to carry out of the room

network	trained band	moved to the other half	cost
dense (lesson 9)	0.8567	**0.4617**	**−0.3950**
convolutional	0.9800	0.9847	+0.0047

Where the defects were, and what it cost





Below chance, and why nothing degraded

- Most students predict a drop that stops well above chance
- The reasoning is sound: shifts usually degrade, they do not vanish
- But this is not a change of **degree** in the inputs
- It is a change of **which** inputs carry the signal
- Knowledge stored per position, at positions never trained

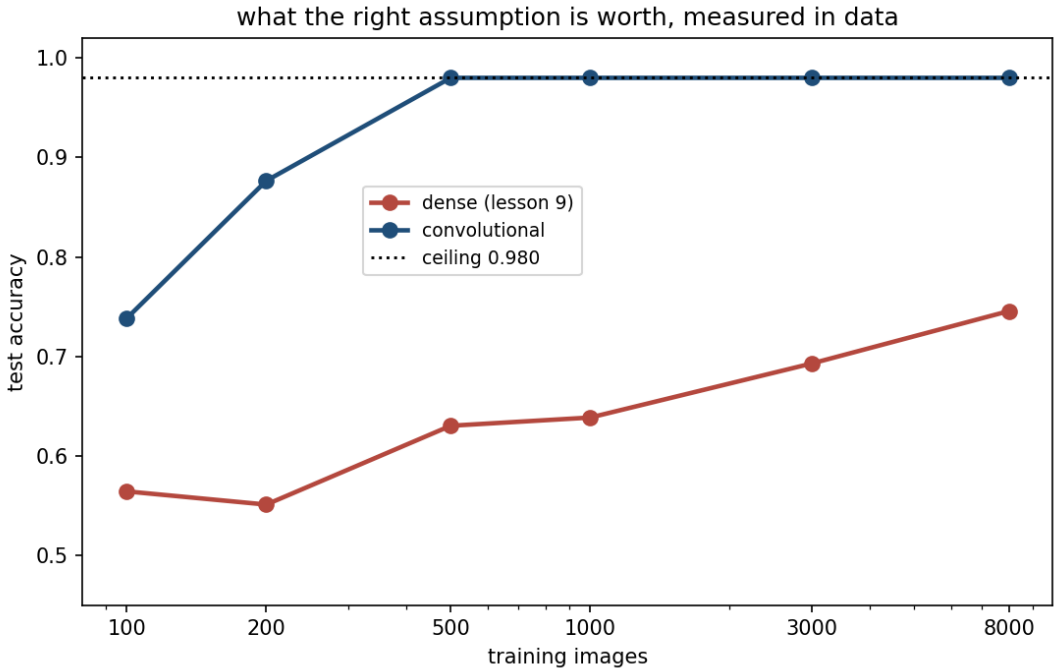
A convolutional layer is a restricted dense layer

- Every function it computes, a dense layer can compute too
- Tie the shared entries, zero the rest: it is strictly **less** expressive
- The functions it gives up all treat row 5 differently from row 15
- Nobody wanted any of those
- **A model that cannot express a wrong answer need not learn to avoid it**

Augmentation buys 8 points and leaves it 44 short

- Show the dense network shifted copies: a moved scratch is still a scratch
- Dense: 0.4617 → **0.5423**, a real gain of **+0.0807**
- Convolutional: 0.9847 → 0.9847, unchanged, with nothing to gain
- Augmentation **teaches** an invariance: approximate, and forgettable
- Architecture **asserts** it: exact, free, permanent

At the ceiling by 500 images; 0.235 short at 8,000



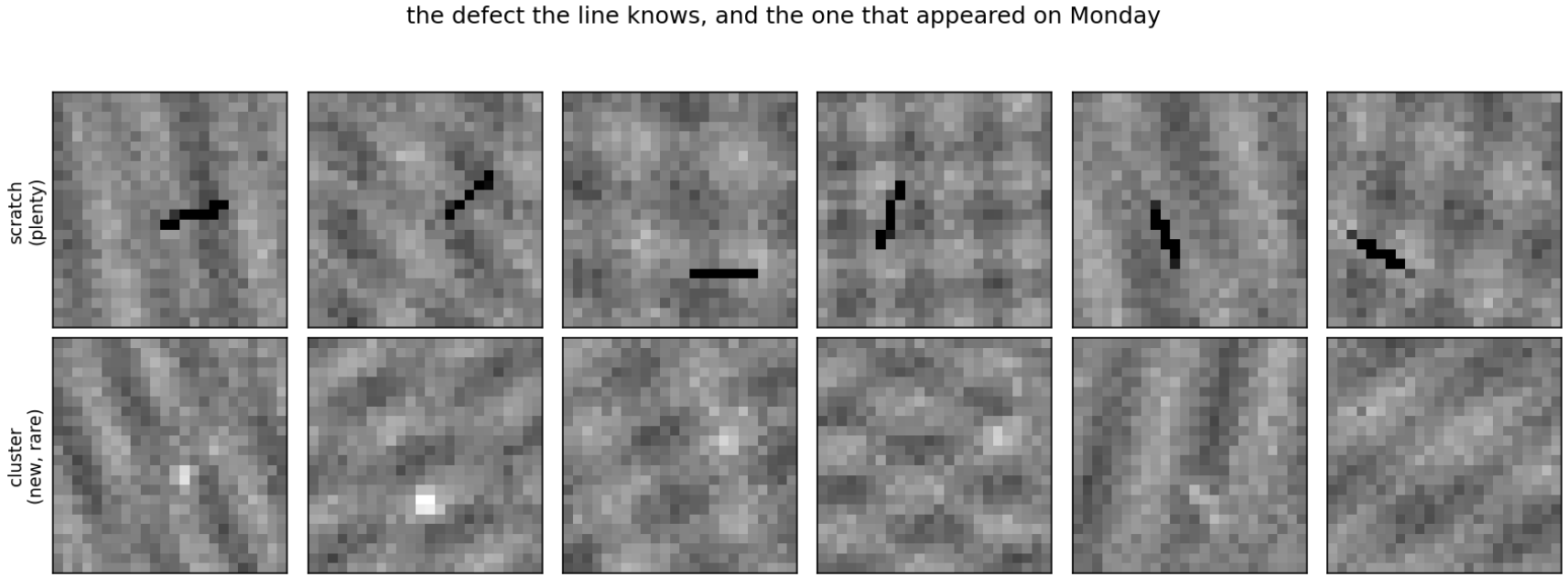
Notebook 2, live

- Both architectures trained head to head, with seed spreads
- The translation experiment, run in front of you
- Augmentation with the epoch budget equalised
- The learning curve from 100 images to 8,000
- The permutation test: hold your prediction

A new defect appears on a Monday

- A contamination **cluster**: several faint specks together
- By Friday, a few dozen labelled photographs
- Section 8 says the network needs several hundred
- Waiting six months is no answer: the line ships now
- Can we borrow from a network trained on something else?

The defect the line knows, and Monday’s



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Start from someone else’s weights

- **Transfer learning**: initialise from a network trained on a related task
- The argument: early layers detect local contrast, which is not scratch-specific
- Two knobs: which **source** task, and which layers stay **frozen**
- Both knobs matter more than the received wisdom suggests

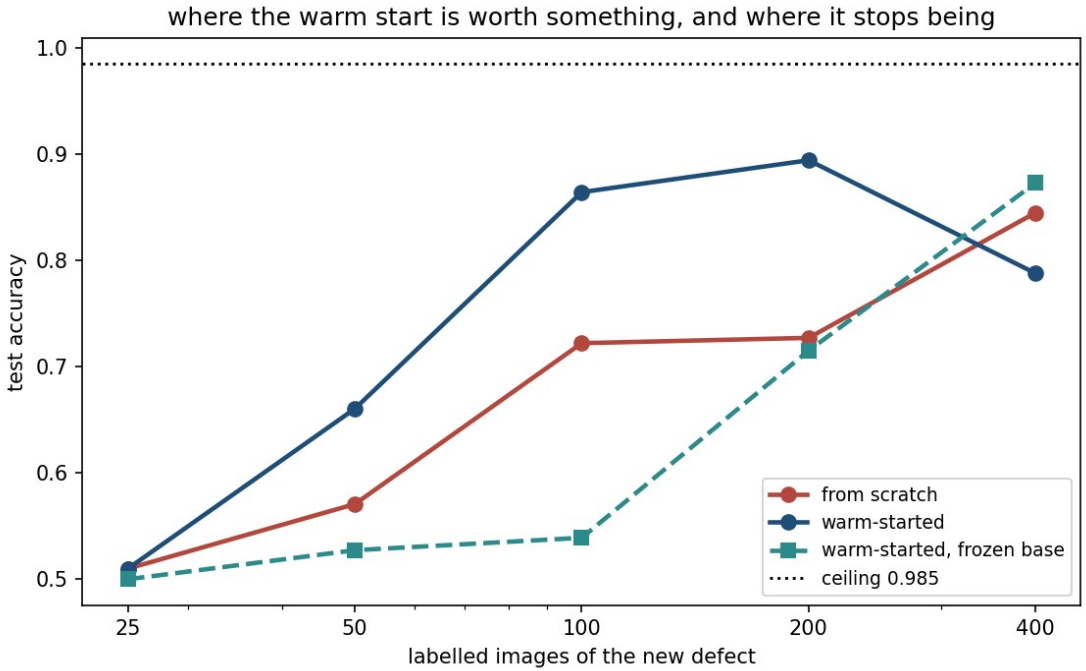
The source task is the whole game

- Pre-train on a task **nobody at Meridian wants solved**
- Naming which of three types a die carries: clean, scratch, particle
- Grading pass/fail teaches **one** detector; typing forces description
- **0.9993** on the source task: a receipt, not a result
- Transfer carries what the source **forced** it to learn

A window, not a slope

labelled images	from scratch	warm-started	gain
25	0.5100	0.5100	0.0000
100	0.7220	**0.8640**	**+0.1420**
200	0.7270	**0.8940**	**+0.1670**
400	0.8447	0.7877	**−0.0570**

The window, drawn



When transfer makes things worse

labelled images	from scratch	frozen, typing source	frozen, scratches only
50	0.5707	0.5273	0.5150
100	**0.7220**	0.5390	0.5047
200	0.7270	0.7150	0.5350

Two rules, and neither is “always pre-train”

- Transfer carries what the source task **forced** the network to learn
- Ask what the source required before trusting its features
- Freezing is a **bet** that the borrowed features are already right
- **In doubt: warm start, and leave everything trainable**

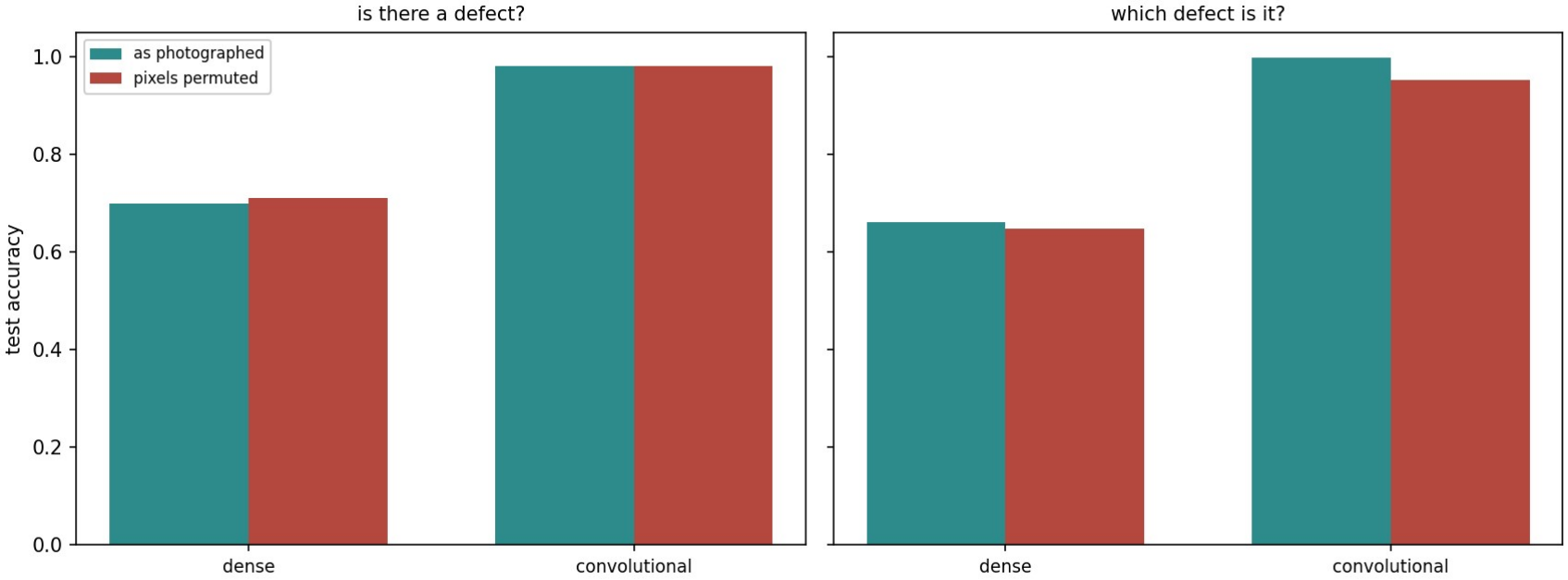
Notebook 3, live

- Pre-training on the task nobody wants solved, up to 0.9993
- The transfer window, measured at five dataset sizes
- Negative transfer, from a source chosen badly on purpose
- Every family the course has met, on one problem

What is the convolution actually using?

- “Images have spatial structure” is **two** claims, not one
- Permute every image’s pixels identically, then train
- Detection: **0.9798 → 0.9800**, no cost at all
- Typing: **0.9970 → 0.9523**, 4.5 points
- The dense network notices neither

Nothing on detection, 4.5 points on typing

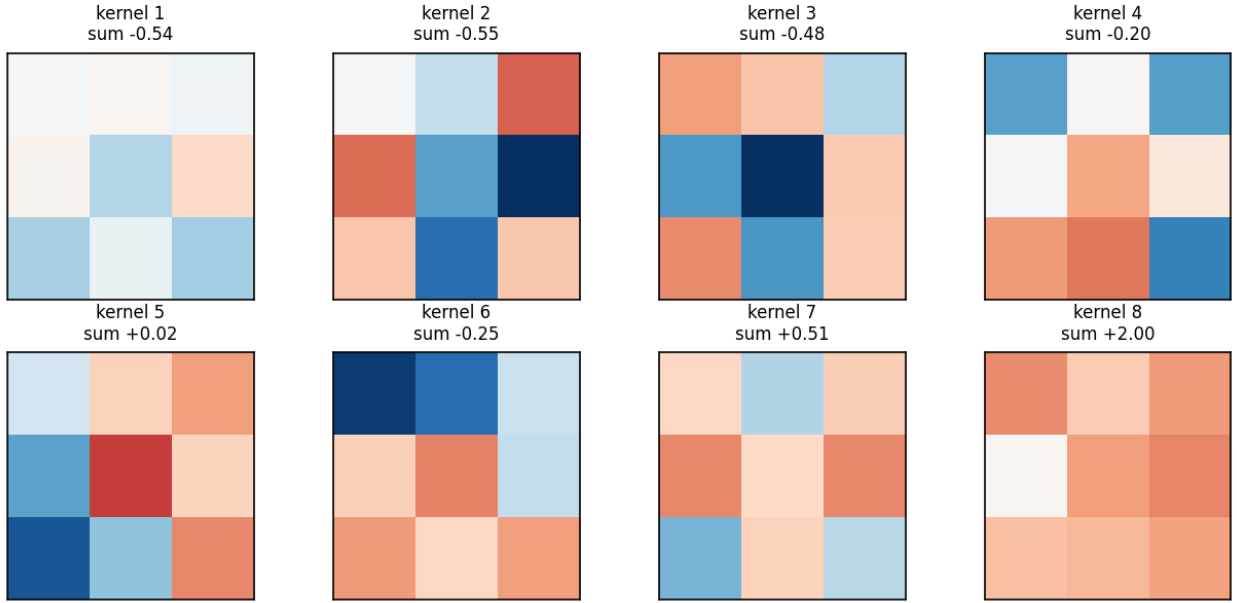


Two assumptions, and only one is about arrangement

assumption	what it says	needed by
weight sharing	a pattern means the same thing wherever it appears	both tasks
locality	nearby pixels belong together	only the typing task

What the first layer learned, unprompted

the eight 3×3 kernels the first layer learned, red positive, blue negative



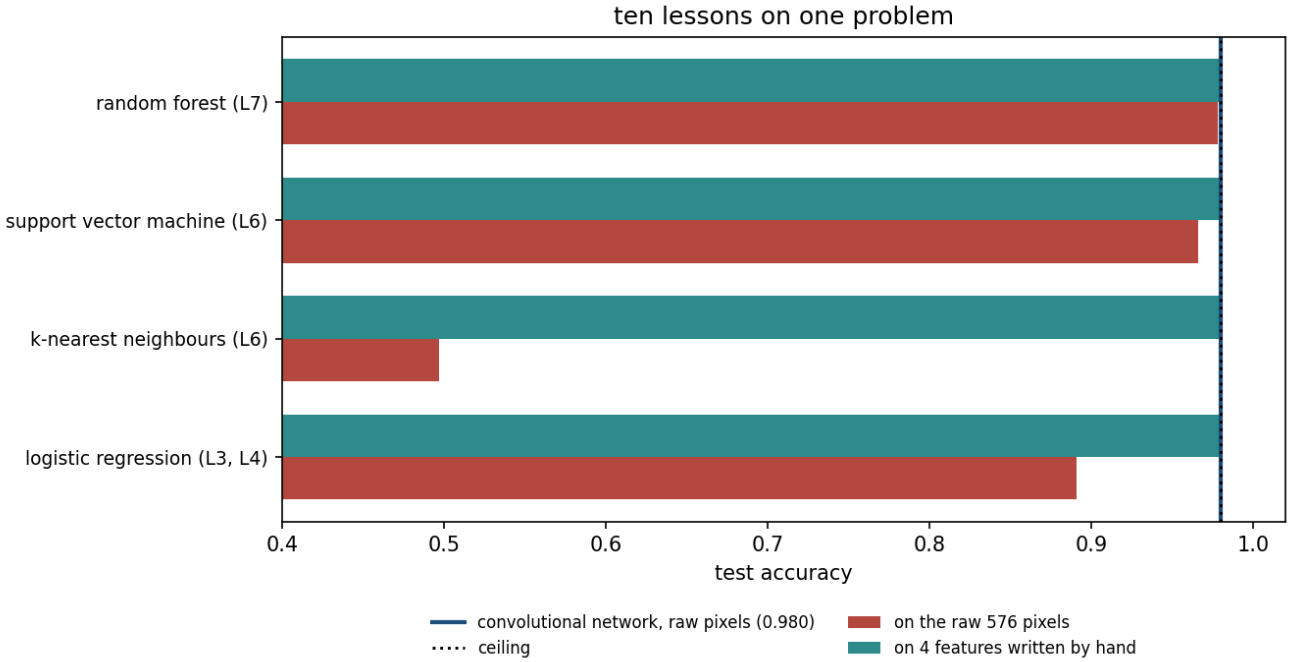
What this lesson did not need

- Today ran on a laptop processor, no graphics card
- Deeper networks revive lesson 9’s gradient problem
- **Batch normalisation** rescales each layer’s activations
- **Residual connections** let the gradient skip a layer
- Neither is needed at 24×24

Ten lessons on one problem

family	raw pixels	4 hand-made features
logistic regression (3, 4)	0.8905	0.9800
k-nearest neighbours (6)	**0.4970**	0.9800
support vector machine (6)	0.9660	0.9800
random forest (7)	**0.9780**	0.9795
dense network (9)	0.7430	-
convolutional network (10)	**0.9800**	-

Four hand-written numbers reach the ceiling



The representation matters more than the model

- A model turns a **representation** into a decision
- Classical methods make you build it; networks learn it, paid for in data
- Lesson 9: a hidden layer worth 39 points on one problem, 0.09 on another
- Today: a random forest within **0.002** of the convolutional network
- **Neither route is a default**

Homework: the last one

- **Exercise 10**, no lesson left to discuss it: work it for the exam
- `Exercises/10_convolutional_networks.md`
- It falls in the week of the **project peer review**: plan for both
- Two reviews to write as well as two to receive

What ten weeks add up to

- **0.8567 to 0.4617** for a dense network, on a defect moved half a die
- The convolutional one went 0.9800 to 0.9847, because it cannot ask *where*
- An assumption that is true beats flexibility that is not
- The representation is doing most of the work; the only question is who builds it
- Thank you, and good luck with the project